ICT

INFORMATION AND COMMUNICATION TECHNOLOGY

GRADE 10



UNIT 9

Database

Student Workbook

"The King of ICT"

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STUDENT DETAILS

Name: _____

School: _____

Institution: _____

Contact: _____

BEFORE WE START

What This Unit Is About

Read this page with your teacher

AT THE END OF THIS UNIT YOU WILL BE ABLE TO

- [ ] Explain a database, and compare manual and electronic databases.
- [ ] Describe the features of a database, such as validity and security.
- [ ] Name the parts of a table and find its degree and cardinality.
- [ ] Find the primary key, composite key and foreign key of a table.
- [ ] Choose data types and create a table in a DBMS.
- [ ] Build one-to-one, one-to-many and many-to-many relationships.
- [ ] Use forms, queries and reports to enter, find and present data.

THREE RULES FOR THE COMPUTER ROOM

- **Always** save your database in your own folder, with your name.
- **Never** type the real phone or ID card numbers of friends - use textbook data.
- **Always** ask your teacher before you delete a table or a relationship.

HOW TO USE THIS BOOK

Each chapter has three parts. First you **read** the chapter with your teacher. Then you **do** the practical task on paper or in a DBMS. Last you **write** the exercises in this book.

You will need: a pencil, an eraser and a ruler for neat tables and arrows, your textbook (pages 269 - 312), and a computer with MS Access 2010 (or LibreOffice Base).

THE WORDS YOU WILL LEARN

Database	Data set	Manual database	Electronic database
Data redundancy	Data consistency	Encryption	Database model
Field	Record	Relational database	Degree
Cardinality	Primary key	Composite key	Foreign key
Data type	DBMS	One-to-one	One-to-many
Many-to-many	Form	Query	Report

"Data is everywhere. Learn to organise it, and you will never lose it again."

- THE KING OF ICT

CHAPTER 1

Databases and Their Features

? TOO MANY PHONE NUMBERS

"I can remember my friends' names, but not all their phone numbers!"

"Write them in a book or save them in your phone - that is a database!"

A **database** keeps data organised, so we can find it quickly. Why is an electronic database better than a paper one?

9.1 What is a database?

We can keep a little data in our memory, but not a large amount. A **database** stores and organises data systematically: a collection of data, organised in a certain manner, is a database. Names and phone numbers written in a book are a **manual database**; saved in a mobile phone they are an **electronic database**.

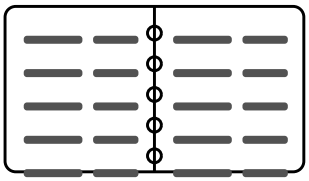




Figure 9.1 - a phone book (manual) and Figure 9.2 - contacts in a phone (electronic)

i DATABASE OR DATA SET?

- A **database** is a collection of logically organised data. Example: a telephone directory.
- A **data set** is a collection of data that is not logically organised. Examples: the data in a word processing document, the data in a spreadsheet.

9.2 Manual and electronic databases

An electronic database saves data in a very small space. Table 9.1 compares the two kinds.

Table 9.1 - a manual database and an electronic database

Manual database	Electronic database
Less efficient	More efficient
Less accurate	More accurate
Less credible	More credible
Difficult to analyse data	Easy to analyse data
The data must stay in the order of input	The data input order can be changed
Difficult to delete unnecessary data	Easy to delete unnecessary data
Difficult to update data	Easy to update data
More space is needed to save data	A very small space is needed to save data
More man power is needed	Less man power is needed
Difficult to share data	Easy to share data
Data cannot be validated	Data can be validated
Access control is difficult	Access control is easy

9.3 Data redundancy and data consistency

A student's name is kept in the admission table and again in the library membership table. Firosa Lai became Firosa Rafeeq, but her name was changed only in the admission table:

Table 9.2 - Admission_table

Student Number	Name	Date of Birth	Date of Admission	Name of the Guardian
100	Manoj Dayarathne	2/9/2008	1/1/2014	G. Dayarathne
101	Firosa Rafeeq	4/2/2008	1/1/2014	K. Fais

Table 9.3 - Library_membership_table

Student Name	Name of the Book	Borrowed Date	Return Date
Firosa Lai	Folk Songs	2/9/2014	16/9/2014
Manoj Dayarathne	Drama	3/9/2014	17/9/2014
Firosa Lai	Basic Maths Concept	3/9/2014	17/9/2014
Manoj Dayarathne	Principles of Media	4/9/2014	18/9/2014

Data redundancy: the same data is stored in several tables.

Data inconsistency: different values for the same data in different tables (Firosa Rafeeq or Firosa Lai?).

A good database keeps redundancy to a **minimum** and keeps data **consistent**.

In a **relational database** each name is written in **only one table**; the other tables use the Student Number and the Book Number (Table 9.4):

Student table

Student Number	Name	Date of Birth	Date of Admission	Name of the Guardian
100	Manoj Dayarathne	2/9/2008	1/1/2014	G. Dayarathne
101	Firosa Rafeeq	4/2/2008	1/1/2014	K. Fais

Borrowed books table

Student Number	Book Number	Borrowed Date	Return Date
101	2222	2/9/2014	16/9/2014
100	3333	3/9/2014	17/9/2014
101	1111	3/9/2014	17/9/2014
100	4444	4/9/2014	18/9/2014

Book table

Book Number	Title	Author's Name
1111	Basic Maths Concept	Sivathasan Ramanathan
2222	Folk Songs	Jayantha Bandara
3333	Drama	Padmamali Menike
4444	Principles of Media	Methuki Thisansa

9.4 More features and database models

- **Increased efficiency:** the tables are well organised, so saving and reading data is efficient.
- **Increased accuracy:** with less redundancy the data stays consistent. In Table 9.4 a name is changed in one place only, so there are fewer mistakes.

	Field Name	Data Type
	Fees	Currency

Field properties

General	Lookup
Format	Currency
Decimal Places	2
Input Mask	
Caption	
Default Value	**1000**
Validation Rule	**>=1000**
Validation Text	
Required	**Yes**
Indexed	No

Figure 9.3 - the field Fees and its properties

Increased validity: field properties are set when the database is planned, and data is checked when it is entered. A fee has 2 decimal places, must be 1000 or more and must be entered (Required: Yes).

Security: a password and encryption stop unauthorised access. **Encryption** translates data so that unauthorised people cannot read it.

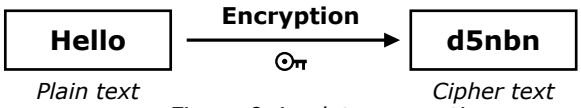


Figure 9.4 - data encryption

Databases are grouped by the **model** used to organise the data. This unit uses the **relational model**.

(i) DATABASE MODELS

- Network model
- Hierarchical model
- Object relational model
- Relational model

(>) DO IT - PRACTICAL 1 (TEXTBOOK PAGES 273 - 274)

1	Your teacher makes a table with the field **Fees** (Currency) in MS Access.
2	Set Decimal Places 2, Default Value 1000 and Validation Rule >=1000.
3	Type **800** as a fee, then **2500**. Which one is accepted? Why?
4	Set Required to Yes and leave a fee empty. Which feature of a database is this?

A Manual or electronic?

8 marks

Is each statement true for a manual or an electronic database? Tick one box.

The statement	Manual	Electronic
Data can be validated.	☐	☐
More man power is needed.	☐	☐
A very small space is needed to save data.	☐	☐
Difficult to delete unnecessary data.	☐	☐
The data input order can be changed.	☐	☐
Access control is difficult.	☐	☐
Easy to share data.	☐	☐
Less credible.	☐	☐

B Find the problem

4 marks

Look at Tables 9.2, 9.3 and 9.4 in section 9.3.

1. Which student names are written more than once in Table 9.3?

.....

2. How many cells of Table 9.3 still show the old name Firosa Lai?

.....

3. Table 9.2 shows Firosa Rafeeq but Table 9.3 shows Firosa Lai. What is this problem called?

.....

4. In the relational database (Table 9.4), in how many tables is the student name stored?

.....

C True or false

6 marks

Write T or F in the box.

1.		A data set is a collection of data that is not logically organised.
2.		A manual database is more efficient than an electronic database.
3.		In a manual database it is difficult to update data.
4.		A good database has a lot of data redundancy.
5.		The relational model is one of the database models.
6.		Encrypted data (cipher text) can be read easily by anyone.

D Match the features

6 marks

Write the correct letter in the box.

A - The feature		B - What it means
1. Minimal data redundancy		**A.** Data is checked at the data entry stage, using the field properties.
2. Data consistency		**B.** The same data is not stored again and again in several tables.
3. Increased efficiency		**C.** Unauthorised access is controlled with a password and encryption.
4. Increased accuracy		**D.** Saving and reading data is fast, because the tables are well organised.
5. Increased validity		**E.** The same data never has different values in different tables.
6. Security		**F.** Fewer mistakes, because a change is made in only one place.

E Field properties

6 marks

Look at Figure 9.3 in section 9.4. Write the answers.

1. Field name: ..... 2. Data type: ..... 3. Decimal places: .....

4. Will the database accept these fees? Write **Yes** or **No** in each box. (3 marks)

Rs. 800 ☐ **Rs. 1000** ☐ **Rs. 2500** ☐

CHAPTER 2

Fields, Records and Keys

? WHICH BOOKS DID FIROSA BORROW?

"How can I find the books that Firosa borrowed from the library?"

"Follow her student number and the book numbers from table to table!"

Tables are made of **fields** and **records**. Special fields called **keys** join the tables.

9.5 Fields, records and tables

- A collection of fields related to one entity (a student, a book) is a **record**.
- A collection of records is a **data table**.
- A collection of related tables is a **relational database**.
- A **field name** names a field. The **data type** tells what kind of data a field can store.

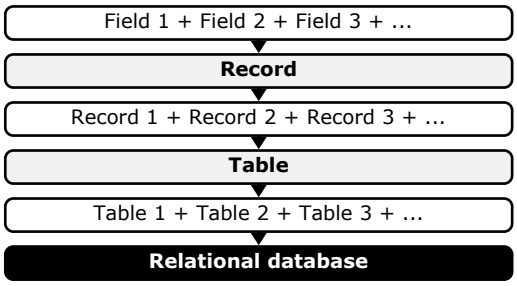


Figure 9.5 - the structure of a relational database

Field (column) ↓

Field names ►	Admission No	Name	Date of Birth	Gender	Phone Number
	1426	Kavindu Prabashwara	2005.05.23	Male	0352287571
Record (row) ►	1427	Meenadevi Ramanathan	2005.08.12	Female	0352235696
	1428	Mohommad Malikkar	2005.02.07	Male	0352815402

Table 9.7 - Student Details

Each **column** is a **field** and each **row** is a **record**. Names can be repeated, but an Admission No is never repeated, so it identifies a record uniquely. The heading row is not a record.

(i) DEGREE AND CARDINALITY

Degree = the number of **fields**.
Cardinality = the number of **records**.
Table 9.7: degree **5**, cardinality **3**.

9.6 The primary key

A column (or a combination of columns) which identifies a record uniquely is the **primary key**. Examples: a National Identity Card Number, a School Admission Number.

(!) A PRIMARY KEY ...

- It should not be empty (**not null**).
- It has **no duplicate** values.

Table 9.8 - Book Table

Accession No	Title	Author's Name	Price of a Book	Number of Books
2131	Let's Access Internet	Mahesh Gankanda	Rs.275.00	10
2132	Computer Studies	Suranimala Basnayaka	Rs.300.00	5
2133	Essay Writing	Nimali Bogoda	Rs.225.00	12
2134	Education Psychology	Dayanath Jayasekara	Rs.400.00	6
2135	Tamil Language Learning	Raj Sudakaran	Rs.225.00	8
2136	Computer Studies	Mohommad Niyas	Rs.150.00	4
2137	Sinhala Tamil Dictionary	Raj Sudakaran	Rs.325.00	6

- In this table: Accession numbers have no duplicates (unique).
- Suranimala Basnayaka and Mohommad Niyas have both written *Computer Studies*.
- Raj Sudakaran has written both *Tamil Language Learning* and *Sinhala Tamil Dictionary*.
- *Essay Writing* and *Tamil Language Learning* have the same price (Rs.225.00).
- *Education Psychology* and *Sinhala Tamil Dictionary* both have 6 copies.

So **Accession No** is the best field for the primary key. The primary key is underlined in the heading of the table (Table 9.9).

9.7 The composite primary key

Table 9.10 - Sports Table

Year	Sport	Captain	Vice Captain
2013	Cricket	Rash Senarathne	Mubharak Husein
2013	Football	Jaliya Saranga	Vinothen Raj
2013	Volleyball	Ahammed Nawas	Sudesh Tharanga
2014	Cricket	Mubharak Husein	Bhanu Dayarathna
2014	Football	Jaliya Saranga	Ahammed Nawas
2014	Volleyball	Bhathiya Gamlath	Sudesh Tharanga

A primary key made of **two or more columns** is a **composite primary key**.

Every column of Table 9.10 has duplicates, but a sport has one captain in a year, so **Year + Sport** is unique.

Each field of the key is underlined.

9.8 The foreign key

A field (or fields) of one table that identifies a row of another table uniquely is a **foreign key**. It connects the two tables. It is the primary key of the other table, and its values must be there.

Table 9.12 - Student Table

Registration Number	Name	Date of Birth	Date of Admission	Name of the Guardian
100	Manoj Dayarathne	2/9/2008	1/1/2014	G. Dayarathne
101	Firosa Rafeeq	4/2/2008	1/1/2014	K. Fais

Primary key of the Student Table

Table 9.13 - Subject Table

Candidate No	Maths	Science	Sinhala	English	Registration No
449683	B	A	C	C	100
449697	A	B	B	A	101

Foreign key of the Subject Table

9.9 Data duplication

Here the marks of two terms are in one table, so the admission number, the name and the date of birth are entered again for each term. Recording column data again and again is **data duplication**.

Table 9.15 - Second term marks

Admission No	Name	Date of Birth	Marks	Term	Year
1426	Kavindu Prabashwara	2005.05.23	69	1	2014
1427	Meenadevi Ramanathan	2005.08.12	82	1	2014
1428	Mohommad Malikkar	2005.02.07	47	1	2014
1426	Kavindu Prabashwara	2005.05.23	79	2	2014
1427	Meenadevi Ramanathan	2005.08.12	68	2	2014
1428	Mohommad Malikkar	2005.02.07	66	2	2014

(!) DISADVANTAGES OF DATA DUPLICATION

- A primary key cannot be selected to keep the identity of records.
- Efficiency is reduced.
- It is difficult to input data.
- It is difficult to update data.
- Data cannot be analysed properly.
- There is a high chance of inaccurate data (Malikkar may become Malakkar).
- Errors can happen when data is deleted.

So the table is split into two tables. **Admission No** is the primary key of the Student Table and the foreign key of the Marks Table. The Marks Table has a composite primary key: **Year + Term + Admission No**.

Table 9.16 - Student Table

Admission No	Name	Date of Birth
1426	Kavindu Prabashwara	2005.05.23
1427	Meenadevi Ramanathan	2005.08.12
1428	Mohommad Malikkar	2005.02.07

Table 9.17 - Marks Table

Year	Term	Marks	Admission No
2014	1	69	1426
2014	1	82	1427
2014	1	47	1428
2014	2	79	1426
2014	2	68	1427
2014	2	66	1428

DO IT - PRACTICAL 2 (TEXTBOOK ACTIVITY, PAGE 278)

1	In Table 9.4 (section 9.3), find the Student Number of Firosa Rafeeq.
2	Find her rows in the Borrowed books table. Write down the book numbers.
3	Find the titles in the Book table. Do the same for Manoj Dayarathne.
4	Underline each primary key. Draw an arrow from each foreign key to its primary key.

A Which key is it?

6 marks

Tick one box for each.

The field (or fields)	Primary key	Composite primary key	Foreign key
Accession No (Book Table 9.8)	☐	☐	☐
Year + Sport (Sports Table 9.10)	☐	☐	☐
Registration No (Subject Table 9.13)	☐	☐	☐
Candidate No (Subject Table 9.13)	☐	☐	☐
Admission No (Marks Table 9.17)	☐	☐	☐
Year + Term + Admission No (Marks Table 9.17)	☐	☐	☐

B Data duplication

6 marks

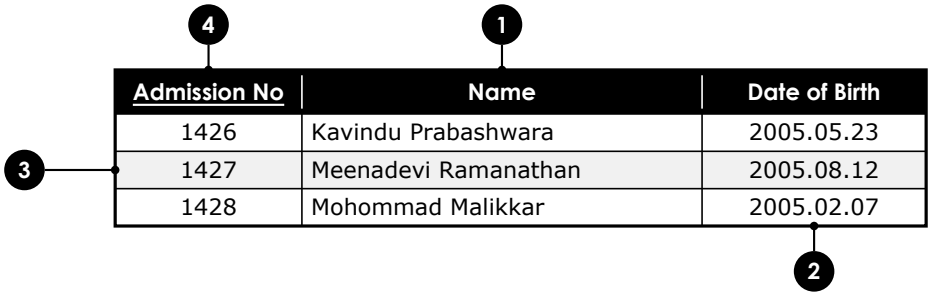
Look at Table 9.15 in section 9.9.

1. How many rows of Table 9.15 show marks above 60? .....
2. How many students are really in this table? .....
3. How many times is the date of birth of each student written? .....
4. Write three disadvantages of data duplication. (3 marks)

C Label the table

6 marks

Use: field name, field, record, primary key. Then write the degree and cardinality.



Admission No	Name	Date of Birth
1426	Kavindu Prabashwara	2005.05.23
1427	Meenadevi Ramanathan	2005.08.12
1428	Mohommad Malikkar	2005.02.07

1. ..... 2. ..... 3. ..... 4. .....

5. Degree: ..... 6. Cardinality: .....

D Degree and cardinality

8 marks

Count the fields and the records of each table in this book.

Table	Degree	Cardinality
Table 9.2 - Admission_table		
Table 9.3 - Library_membership_table		
Table 9.8 - Book Table		
Table 9.10 - Sports Table		

E Choose the correct answer

6 marks

Underline your answer.

1. A table has 6 fields and 9 records. Its degree is
   (a) 6 (b) 9 (c) 15 (d) 54
2. The cardinality of the Book Table (Table 9.8) is
   (a) 5 (b) 6 (c) 7 (d) 8
3. The best primary key for a table of students is
   (a) Name (b) Gender (c) Admission No (d) Phone Number
4. A primary key must be
   (a) empty (b) repeated (c) unique (d) a name
5. The composite primary key of Table 9.10 is
   (a) Year (b) Sport (c) Year + Sport (d) Captain
6. Recording the same column data again and again is data
   (a) duplication (b) validation (c) encryption (d) security

"Be like a primary key: one of a kind, and never empty."

- THE KING OF ICT

CHAPTER 3

Creating a Relational Database

? FROM PAPER TO COMPUTER

"Our library writes every new book in a big register. Can a computer do it?"

"Yes! We can make a database in MS Access or LibreOffice Base."

Choose a **data type** for every field, create the tables in a **DBMS**, and then join the tables with **relationships**.

9.10 Data types

The type of data related to one field is its **data type**. Choose it when the table is created.

Data type	What it stores	Examples
Numeric - Integer	plus or minus whole numbers	12, -23
Numeric - Real	plus or minus numbers with decimal places	8.125, -2.64, 4.00
Text	letters, numbers or special characters (%, *, -); not used for calculations	NIC number 889534731V, a name, an address, telephone number 0112785123
Currency	money values	$12.45, Rs.35.00
Date / Time	dates and times	12/23/2013, 7.35 a.m.
Boolean	only two (logical) values: True / False or Yes / No	True, No

The **size** of a field can be changed to suit its data type and the number of characters. Example: Subject Name - 15 characters, Subject Code - 5 characters.

9.11 Database Management Systems

A collection of objects which helps the user to plan and maintain a database is a **Database Management System (DBMS)**.

Its objects are **tables**, **queries**, **forms** and **reports** (Figure 9.6). This unit uses **Microsoft Office Access** and **LibreOffice Base**.

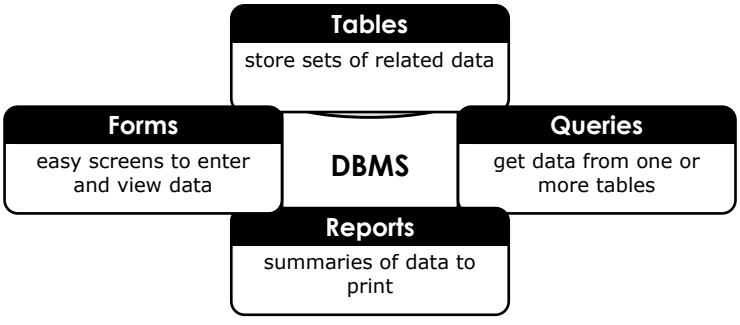


Figure 9.6 - the objects of a DBMS

Table 9.18 - some DBMS software and the companies that make them

Software	Made by	Free and open source / Proprietary
Access	Microsoft	Proprietary
Base	The Document Foundation	Free and open source
Oracle	Oracle Corporation	Proprietary
MySQL	Oracle Corporation	Free and open source

9.12 Creating a table in a DBMS

A table is planned in **Design View** and the data is typed in **Datasheet View**. (The commands to start the software can be different in other operating systems.)

Step	Microsoft Office Access	LibreOffice Base
Start the software	Start → Programs → Microsoft Office → Microsoft Office Access → Blank Database → select a folder and type a name → Create	Start → Programs → LibreOffice → LibreOffice Base → Create a new database → Next → Finish → select a folder and type a name → Save
Plan the table	Create → Table Design → type the field names and data types → set the field properties → save the table	Table in Design View → type the field names and data types → set the field properties → save the table
Make the primary key	select the row(s) of the key → click Primary Key → save the table	right click the row of the field → Primary Key → save the table
Enter the data	select the table in 'All Access Objects' → Datasheet View → type the data → close the table	double click the table under Tables (or right click, Open) → type the data → close the table

Library

	Field Name	Data Type
🔑	Accession_Number	Text
	Title	Text
	Author	Text
	Price	Currency
	Number_of_books	Number

Figure 9.8 - the Library table in Design View

The **key** icon shows the primary key, **Accession_Number**. The price is **Currency** and the number of books is **Number**.

An accession number or a telephone number is **Text**: it is never used for calculations.

DO IT - PRACTICAL 3 (TEXTBOOK ACTIVITY, PAGE 285; PAGES 287 - 289)

1	On paper, list **10 books** of your school library under the fields of Table 9.8.
2	Start MS Access (or LibreOffice Base). Make a blank database with your name.
3	In Design View, make the **Library** table of Figure 9.8 with its primary key.
4	Open the table in Datasheet View, type your 10 books, then save and close it.

9.13 The one-to-one relationship

If one row of a table is related to only **one** row of another table (and that row to only one row of the first), the relationship is **one-to-one**. Table 9.19 (page 289) has Grade 5 scholarship marks. It is split into two tables:

Table 9.20 - Marks

Index No	Marks	Admission No
23234	151	1426
23876	186	1427
23758	172	1428
23765	193	1429

Table 9.21 - Students Table

Admission No	Name	Date of Birth
1426	Kavindu Prabashwara	2005.05.23
1427	Meenadevi Ramanathan	2005.08.12
1428	Mohommad Malikkar	2005.02.07
1429	Rashmi Janodara	2005.06.16

Admission No is the primary key of the Students Table and the foreign key of the Marks table, whose primary key is **Index No**. The primary key of one table is entered as a foreign key in the other table. An arrow is drawn from the foreign key to the primary key.

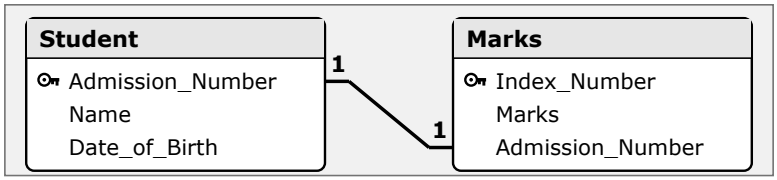


Figure 9.10 - a one-to-one relationship

9.14 The one-to-many relationship

If one record of a table is related to **several** records of the other table, the relationship is **one-to-many**. If B is the related table:

Table 9.22 - the features of a one-to-many relationship

Table A (the "one" table)	Table B (the related "many" table)
A record of Table A can be related to several records of Table B.	Each record of Table B is related to only one record of Table A.
Some records of Table A may not be related to any record of Table B.	Every record of Table B must be related to a record of Table A.
Data in the primary key column cannot be duplicated.	Data in the foreign key column can be duplicated.

Table 9.23 (page 291) kept the fees in the student table: the data of 1426 was duplicated and 1427 had an incomplete record. Split into two tables, one student can have **0, 1, 2 or 3** fee records:

Table 9.25 - Facility Fees Table

Receipt Number	Payment Date	Fees	Admission No
1000	2014.01.04	40.00	1426
1001	2014.01.06	120.00	1428
1002	2014.01.12	80.00	1426

Table 9.24 - Students Table

Admission No	Name	Phone Number
1426	Kavindu Prabashwara	0352287571
1427	Meenadevi Ramanathan	0352235696
1428	Mohommad Malikkar	0352815402

To make a one-to-many relationship, the primary key of the "one" table is entered as a foreign key in the related ("many") table. MS Access shows **1** and the many sign.

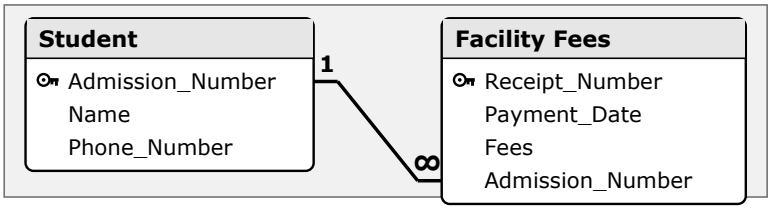


Figure 9.11 - a one-to-many relationship

9.15 The many-to-many relationship

If a record of one table is related to several records of another table, **and** a record of that table is related to several records of the first, the relationship is **many-to-many**:

Table 9.26 - Students who do sports

Admission No	Name	Date of Birth	Sport No	Sport	Team	Number of permanent members
1426	Kavindu Prabashwara	2005.05.23	S001	Cricket	A	11
1426	Kavindu Prabashwara	2005.05.23	S002	Football	A	16
1428	Mohommad Malikkar	2005.02.07	S001	Cricket	B	11
1428	Mohommad Malikkar	2005.02.07	S002	Football	A	16
1428	Mohommad Malikkar	2005.02.07	S003	Volleyball	B	6
1429	Janith Asheka	2005.04.15	S001	Cricket	A	11
1429	Janith Asheka	2005.04.15	S003	Volleyball	A	6

It is changed into **two one-to-many** relationships with a **new table**. The primary keys of the Student Table and the Sports Table go into the new Student_Sport Table as foreign keys. Its primary key is **Admission No + Sport No**.

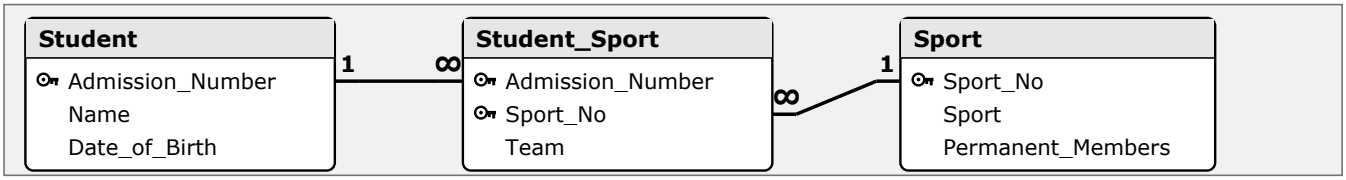


Figure 9.12 - a many-to-many relationship as two one-to-many relationships

9.16 Relationships in a DBMS

CREATING A RELATIONAL DATABASE

1. Design and create the table.
2. Assign the primary key.
3. Create relationships between the tables.
4. Design forms for data entry.
5. Enter the data.
6. Retrieve information using queries.

MS Access: Database Tools → Relationships → Show Table → select a table → Add → drag the primary key to the foreign key → tick 'Enforce Referential Integrity' → 'Cascade Update Related Fields' and 'Cascade Delete Related Records' → Create

LibreOffice Base: Tools → Relationships → select a table → Add → drag the primary key to the foreign key → right click the line → Edit → 'Update cascade' and 'Delete cascade' → OK

With **Referential Integrity**, a changed primary key value changes the foreign keys too, and deleting a student deletes his or her related records. Enter the students first; then click **+** in front of a student to enter the marks.

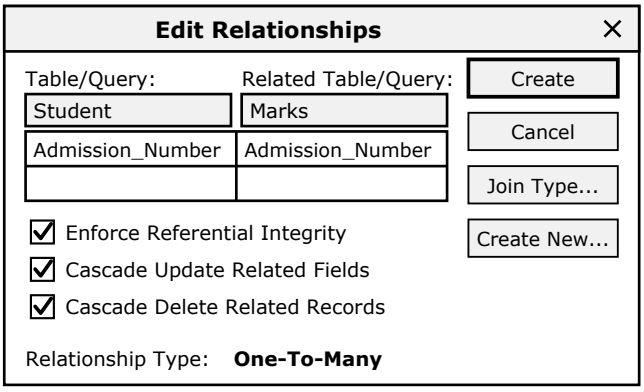


Figure 9.16 - Edit Relationships (Access)

DO IT - PRACTICAL 4 (TEXTBOOK PAGES 297 - 299)

1	Make the **Student** table (Figure 9.13) and the **Marks** table (Figure 9.14).
2	Database Tools → Relationships → Show Table: add both tables.
3	Drag **Admission_Number** from Student to Marks, tick the three boxes, Create.
4	Type the students first. Then click **+** in front of a student to add marks.

"Choose the right type for your data, and the right road for your life."

- THE KING OF ICT



Colour me!

A Choose the data type

7 marks

Textbook Activity, page 285. Write the most suitable data type.

Field	Data type
Admission Number	
Date of Admission	
Subject Code	
Doctor's Fee	
Department Name	
Number of Passengers	
Is Registered?	

B Which data type?

8 marks

Tick the data type of each value.

Value	Integer	Real	Text	Currency	Date/ Time	Boolean
-23	☐	☐	☐	☐	☐	☐
8.125	☐	☐	☐	☐	☐	☐
Rs.35.00	☐	☐	☐	☐	☐	☐
7.35 a.m.	☐	☐	☐	☐	☐	☐
True	☐	☐	☐	☐	☐	☐
NIC number 889534731V	☐	☐	☐	☐	☐	☐
Telephone no. 0112785123	☐	☐	☐	☐	☐	☐
-2.64	☐	☐	☐	☐	☐	☐

C Which relationship?

6 marks

Tick one box for each pair.

The two things	One-to-one	One-to-many	Many-to-many
A student and his or her scholarship Index No	☐	☐	☐
A student and the fee receipts he or she gets	☐	☐	☐
Students and the sports they play	☐	☐	☐
A student and his or her term marks	☐	☐	☐
A citizen and his or her NIC number	☐	☐	☐
Students and the library books they borrow	☐	☐	☐

"Strong relationships make strong tables - and strong friendships."

- THE KING OF ICT

D Complete the Student_Sport Table

7 marks

Use Table 9.26 in section 9.15. 1 mark for each row.

Student_Sport Table

Admission No	Sport No	Team

Write one row for each student in each sport: the Admission No, the Sport No and the Team.

The primary key is Admission No + Sport No, so no row is repeated.

E Count the related records

6 marks

Use the fee tables of 9.14 and Table 9.26 of 9.15.

1. How many times did student 1426 pay fees?

2. How many times did student 1427 pay fees?

3. What is the total of the fees paid by 1426?

4. How many students play cricket (S001)?

5. How many sports does student 1428 play?

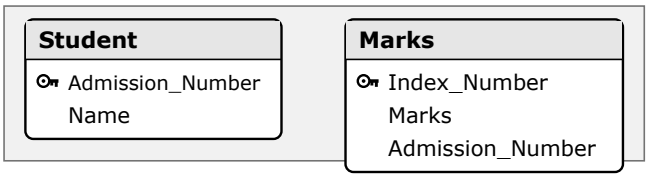
6. In which team does 1428 play volleyball (S003)?

F Draw the relationships

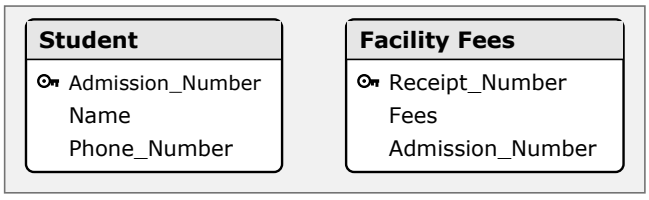
6 marks

Join the related fields with a line. Write 1 or the many sign at each end.

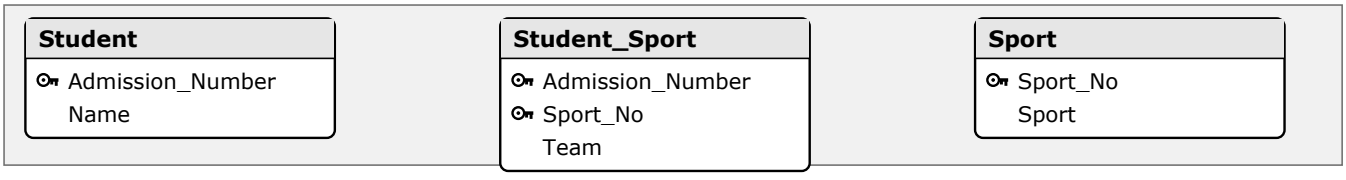
1. one-to-one



2. one-to-many

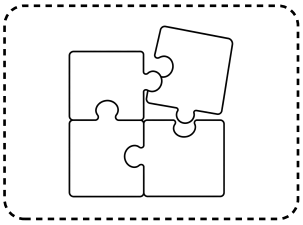


3. many-to-many (2 marks)



"Save your work often. Save your time always."

- THE KING OF ICT



Colour me!

CHAPTER 4

Forms, Queries and Reports

? WHO GOT THE TOP MARK?

"Can the computer show only the 2014 Maths marks, and print them?"

"Yes! A query finds the data, and a report prints it."

Forms enter and show data, **queries** get information from the tables, and **reports** present it on paper.

9.17 Forms

Forms are used to enter data into a table and to display the data. Each record entered through a form goes to the last row of the table.

Admission_Number	001
Surname_with_Initials	Karunajeewa R.J.C.
Date_of_Birth	8/10/2003
Telephone_Number	0715648453
Admission_Fees	Rs. 560.00

Add New Record **Delete Record**

Figure 9.22 - a record of the Student table in a form

To ...	In MS Access
Build a data form	Create → Form Design → Add Existing Fields → Show tables → select the table → drag the fields to the form
Show the records	Home → View → Form View
Add a button	Design → 'Control Wizards' → Button → draw it on the form → Record Operations → Add New Record → give a name → Finish

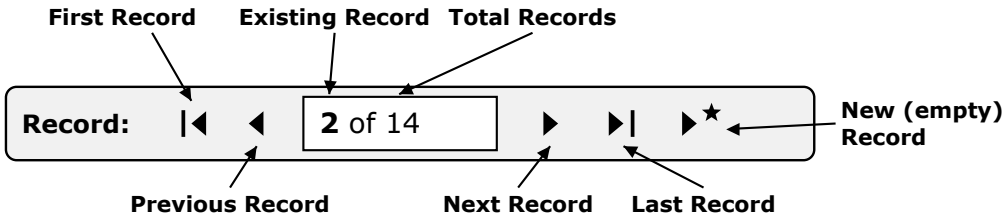


Figure 9.20 - the Record Navigation Bar moves from record to record

9.18 Queries

A **query** gets the information we need from the data of related tables. It shows the data of selected fields. Here Name, Subject and Marks are taken from three tables:

Students

Student No	Name
A1	Pubudu
A2	Faris
A3	Kamalan
A4	Thanuja

Marks

Student No	Subject No	Marks
A1	B1	67
A2	B3	89
A1	B3	50
A3	B2	75

Subjects

Subject No	Subject
B1	Maths
B2	Science
B3	English

→

The query result

Name	Subject	Marks
Pubudu	Maths	67
Faris	English	89
Pubudu	English	50
Kamalan	Science	75

Figure 9.25 - getting data with a query

Design a query: Create → Query Design → Show Table: select the tables, Add → select the fields → Run. To see only the marks of 2014, type **2014** in the **Criteria** row of the Year field:

Field:	Admission_Number	Surname_with_Initials	Maths	Year
Table:	Student	Student	Marks	Marks
Sort:				
Show:	☑	☑	☑	☑
Criteria:				**"2014"**
or:				

Figure 9.29 - the query design grid with a Criteria

Figure 9.30 - the result with the Criteria "2014"

Admission_Number	Surname_with_Initials	Maths	Year
001	Karunajeewa R.J.C.	56	2014
002	Somarathna A.B.G.	66	2014
003	Rasaputhram S.N.	68	2014
004	Jayasena L.K.R.	92	2014
005	Malikkar M.N.M.	76	2014
006	Radhakrisna T.U.	43	2014
007	Faslina M.G.	80	2014

Calculations: a query can calculate with fields. To add the three marks of each student, add a new column named **Total**. Type the expression below, or right click the column and choose **Build** to use the Expression Builder.

Total: [Marks]![Maths]+[Marks]![Science]+[Marks]![English]

9.19 Reports

Reports are used to get printed copies of important information. Both tables and queries can be used. **Report Wizard:** Create → Report Wizard → select the table or query → select the fields → Next ... → Finish.

Fees

Admission_Number	Surname_with_Initials	Admission_Fees
001	Karunajeewa R.J.C.	Rs. 560.00
002	Somarathna A.B.G.	Rs. 560.00
003	Rasaputhram S.N.	Rs. 560.00
004	Jayasena L.K.R.	Rs. 560.00
005	Malikkar M.N.M.	Rs. 560.00
006	Radhakrisna T.U.	Rs. 560.00
007	Faslina M.G.	Rs. 560.00
	Total	**Rs. 3,920.00**

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Figure 9.40 - a report with a total

A total: in Design View, use the text box tool (**ab**) to put a text box in the **Report Footer**. It is **Unbound** - no field is given to it.

In the Property Sheet click the **...** button of Control Source and type **=Sum([Student Table]![Admission Fees])** in the Expression Builder.

Print: Report View → Print Preview → Print.

DO IT - PRACTICAL 5 (TEXTBOOK PAGES 300 - 311)

1	Make a form for the Student table with an **Add New Record** button.
2	Make a query of Student and Marks with the Criteria **2014** under Year. Run it.
3	Add the **Total** column to a query of the 2013 marks. Run it.
4	Make a report with the Report Wizard, add the fees total and print it.

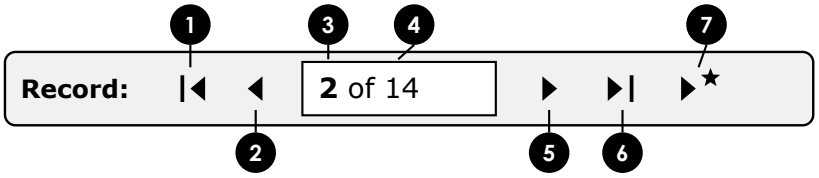
"Ask the right question, and the query gives the right answer."

- THE KING OF ICT

A Label the navigation bar

7 marks

Write the name of each button or box.



1. ..... 2. ..... 3. ..... 4. .....

5. ..... 6. ..... 7. .....

B Run the query

6 marks

Use the three tables of Figure 9.25 in section 9.18. 2 marks for each row.

Fields: Name, Subject, Marks

Criteria of Marks: **>60**

Write the records that the query shows.

The query result

Name	Subject	Marks

C Calculate the totals

8 marks

The 2013 marks (Figure 9.33). Total = Maths + Science + English.

Surname_with_Initials	Maths	Science	English	Total
Karunajeewa R.J.C.	67	45	78	
Somarathna A.B.G.	76	81	55	
Rasaputhram S.N.	37	54	35	
Jayasena L.K.R.	87	72	68	
Malikkar M.N.M.	62	56	43	
Radhakrisna T.U.	32	45	65	
Faslina M.G.	70	89	65	

Write the total of each student. (7 marks)

Who has the highest total? (1 mark)

.....

D Table, query, form or report?

8 marks

Which object does each job? Tick one box.

The job	Table	Query	Form	Report
Stores the data in rows and columns.	☐	☐	☐	☐
Shows only the marks of 2014.	☐	☐	☐	☐
A screen to enter one record at a time.	☐	☐	☐	☐
Prints the marks for the notice board.	☐	☐	☐	☐
Calculates a Total column for each student.	☐	☐	☐	☐
Has a Record Navigation Bar.	☐	☐	☐	☐
Shows the total of the fees in its footer.	☐	☐	☐	☐
Is planned with field names and data types.	☐	☐	☐	☐

E The report total

4 marks

Look at section 9.19 and Figure 9.40.

1. Where is the text box for the total placed?

2. Why is the new text box 'Unbound'?

3. Write the expression for the total of the admission fees.

4. Seven students paid Rs. 560.00 each. What total does the report show?

F Answer these

4 marks

Write full sentences.

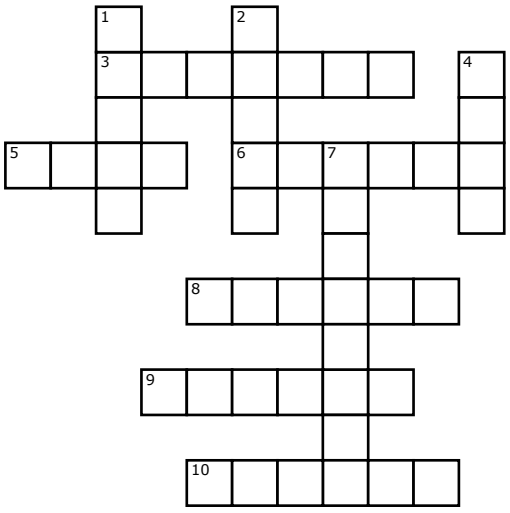
1. What is the difference between a form and a report? (2 marks)

2. Write the expression of a Total column that adds the Maths, Science and English marks. (2 marks)

G Crossword

10 marks

All the words are in this unit. Write one letter in each square.



ACROSS

3. A text box with no field given to it is ... (7)
5. It is used to enter and display data, one record at a time (4)
6. The DBMS made by Microsoft (6)
8. The total of the fees goes in the Report ... (6)
9. It gives a printed copy of information (6)
10. The Report ... helps you to design a report (6)

DOWN

1. It gets information from the data of related tables (5)
2. The new column that adds Maths, Science and English (5)
4. The DBMS of LibreOffice (4)
7. Typing 2014 under Year in a query gives the ... (8)

"Curious minds ask questions. Clever minds find the answers."

- THE KING OF ICT

REVISION

Unit 9 in One Page

Read this the night before the test

Topic	The key facts
Database	A collection of logically organised data (a telephone directory). A data set is not logically organised (the data in a word processing document).
Manual / electronic	Kept in a book / on an electronic medium. Electronic: more efficient, accurate and credible; easy to update, share and validate; very little space.
Features	Minimal data redundancy, data consistency, increased efficiency, accuracy and validity, security (a password and encryption).
Problems	Redundancy: the same data in several tables. Inconsistency: different values for the same data.
Models	Network, hierarchical, object relational and relational models.
Table	Field = column, record = row. A collection of related tables is a relational database.
Degree, cardinality	Degree = the number of fields. Cardinality = the number of records.
Primary key	Identifies a record uniquely: not null, no duplicates. It is underlined.
Composite key	A primary key of two or more fields (Year + Sport). Underline each field.
Foreign key	A field that is the primary key of the other table. It connects two tables.
Data types	Number (integer, real), Text, Currency, Date/Time, Boolean (Yes/No).
DBMS	Objects: tables, queries, forms, reports. Access, Base, Oracle, MySQL.
Relationships	One-to-one, one-to-many (1 and many), many-to-many (needs a new table).
Forms, queries, reports	Enter and display data / get information (Criteria, a Total column) / print information (=Sum in the Report Footer).

★ TICK WHEN YOU CAN DO IT

- ☐ I can explain a database and compare manual and electronic databases.
- ☐ I can name the six features of a database.
- ☐ I can find the degree and the cardinality of a table.
- ☐ I can find the primary key, the composite key and the foreign key.
- ☐ I can choose data types and create a table in a DBMS.
- ☐ I can draw one-to-one, one-to-many and many-to-many relationships.
- ☐ I can make a form, a query with criteria and a report.

"Read each question twice. Check each answer once more."

- THE KING OF ICT

WORD POWER

Unit 9 Word List

Twenty-seven words you must know

Word	What it means
Database	A collection of logically organised data.
Data set	A collection of data that is not logically organised.
Manual database	Data written by hand, e.g. in a phone book.
Electronic database	Data kept on an electronic medium, e.g. in a mobile phone or a DBMS.
Data redundancy	The same data stored in several tables.
Data inconsistency	Different values for the same data in different tables.
Encryption	Translating data so that unauthorised people cannot read it.
Database model	The way data is organised: network, hierarchical, object relational, relational.
Field	A column of a table. It has a field name and a data type.
Record	A row of a table: the fields related to one entity.
Table	A collection of records.
Relational database	A collection of related tables.
Degree	The number of fields in a table.
Cardinality	The number of records in a table.
Primary key	A column (or columns) that identifies a record uniquely. Not null, no duplicates.
Composite key	A primary key made of two or more columns, e.g. Year + Sport.
Foreign key	A field that is the primary key of another table. It connects the tables.
Data duplication	Recording the same column data again and again.
Data type	The type of data of a field: Number, Text, Currency, Date/Time, Boolean.
DBMS	Database Management System: tables, queries, forms and reports.
One-to-many	One record of a table is related to several records of another table.
Many-to-many	Records on both sides are related to several records. It needs a new table.
Referential integrity	Keeps related tables correct: cascade update and cascade delete.
Form	A screen to enter and display data.
Query	Gets information from selected fields of related tables.
Criteria	What is typed in a query to choose the records, e.g. 2014 under Year.
Report	Presents information as a printed copy.

"Every word you learn today is a new record in the database of your mind."

- THE KING OF ICT

END OF UNIT

Unit 9 Test

Time: 45 minutes Total: 40 marks

Name _____ **Class** _____ **Marks** ____

1 Choose the correct answer

8 marks

Underline your answer.

1. A collection of logically organised data is a
   (a) data set (b) database (c) spreadsheet (d) field
2. Which of these is free and open source DBMS software?
   (a) Access (b) Oracle (c) Base (d) MS Word
3. The number of records in a table is its
   (a) degree (b) cardinality (c) field (d) key
4. The best data type for the field Price of Book is
   (a) Text (b) Boolean (c) Currency (d) Date/Time
5. Which object is used to get printed copies of information?
   (a) Table (b) Form (c) Query (d) Report
6. What is the relationship between students and the sports they play?
   (a) one-to-one (b) one-to-many (c) many-to-many (d) none
7. A many-to-many relationship is changed into two one-to-many relationships with a new
   (a) form (b) report (c) table (d) query
8. A - A primary key cannot be empty. B - A foreign key builds a connection between two tables. C - Data in a foreign key column can never be duplicated. Which are correct?
   (a) A and B only (b) B and C only (c) A and C only (d) A, B and C

2 Fill in the blanks

8 marks

One word in each blank.

1. Different values for the same data in two tables is data ..... .
2. Changing data so that others cannot read it is called ..... .
3. In a table, a column represents one ..... .
4. A primary key made of two or more columns is a ..... key.
5. The data type with only True and False is ..... .
6. In MS Access, a table is planned in ..... View.
7. Typing 2014 under Year in a query gives the ..... .
8. =Sum( ) is typed in the Expression ..... .

3 Look at the tables

8 marks

Name the parts 1 - 4. Then answer 5 - 8.

1

Table 9.24 - Students Table

Admission No	Name	Phone Number
1426	Kavindu Prabashwara	0352287571
1427	Meenadevi Ramanathan	0352235696
1428	Mohommad Malikkar	0352815402

2

3

Table 9.25 - Facility Fees Table

Receipt Number	Payment Date	Fees	Admission No
1000	2014.01.04	40.00	1426
1001	2014.01.06	120.00	1428
1002	2014.01.12	80.00	1426

4

1. ..... 2. ..... 3. ..... 4. .....

5. The degree of the Facility Fees Table:

.....

6. The cardinality of the Students Table:

.....

7. The relationship between the two tables:

.....

8. How many times did student 1426 pay fees?

.....

4 Answer these

16 marks

Write in the space given.

1. What is a primary key? Write two of its special attributes. (3 marks)

2. Write the most suitable data type for these fields of the Library table: Title, Price, Number of Books, Registration status. (4 marks)

3. Students and sports have a many-to-many relationship. Write the field names of the three tables that are needed. Underline the primary key of each table. (3 marks)

4. Write two differences between a manual and an electronic database. (2 marks)

5. What are forms used for? What are reports used for? (2 marks)

6. Write the expression for the total of the fees in the Report Footer. (2 marks)